BI / read / 20

BI 1
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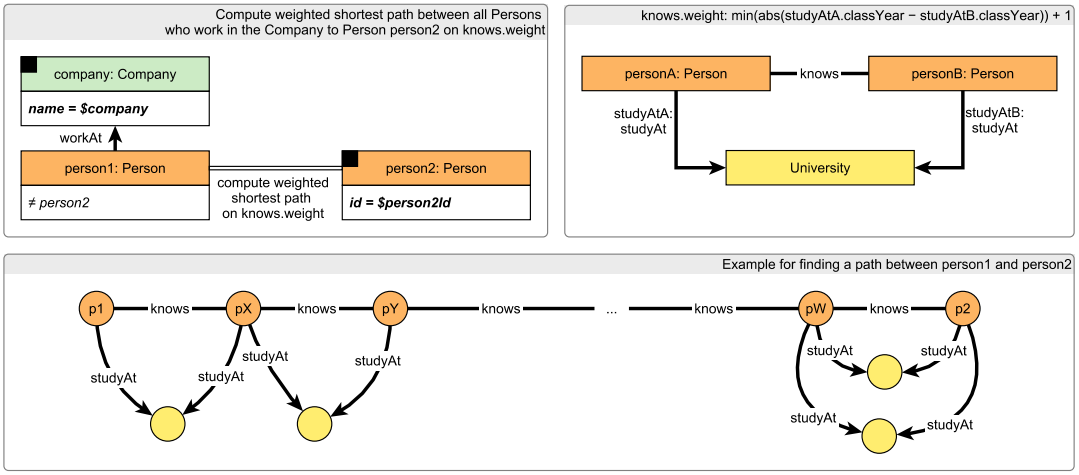
query	BI / read / 20
title	Recruitment
pattern	
description	Consider knows edges where the endpoint Persons attended the same University and set the weight of the edge to the absolute difference between the year of enrolment plus 1. If the Persons attended multiple universities, we select the smallest (`min`) value. Formally: $$w = \min_{\text{studyAt}_A,\text{studyAt}_B} \lvert \text{studyAt}_A.\text{classYear} - \text{studyAt}_B.\text{classYear} \rvert + 1$$ Given a `$company` and a Person `person2` with ID `$person2Id` (who is not working and has not worked at `$company`), find a different Person (`person1`) who works or at some point worked in `$company` and is reachable from `person2` through people who have studied together through the shortest weighted path. If there are multiple Person `person1` nodes with the same shortest path length, return all of them.
params	1 | `$company` \| Long String \| Companies with a similar number of employees (former or current) are selected 2 \| `$person2Id` | ID | (a) There is guaranteed to be no path between any `person1` working at `company` and `person2` (b) There is guaranteed to be a 2-hop path between at least one `person1` working at `company` and `person`
result	1 | `person1.id` | ID | R 2 | `totalWeight` | 32-bit Integer | C
sort	1 | `totalWeight` | ↑ 2 | `person1.id` | ↑
limit	20
CPs	3.3, 7.6, 7.7, 7.8, 8.4, 8.6
relevance	To find the weighted shortest paths efficiently, the system can use e.g. a bidirectional Dijkstra algorithm. As the edge weights do not depend on any parameter, systems can pre-compute them (if they do not interleave reads and writes).